

Rui Wang

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Education

Ph.D. in Cyberspace Security, Harbin Institute of Technology 2026.08-2030.07

M.Sc. in Electrical and Electronic Engineering, The University of Hong Kong 2025.08-2026.07

Main Courses: Pattern Recognition and Machine Learning (A), Digital Image Processing and Computer Vision (A-), Digital Signal Processing (A-), Biomedical Signals and Systems (B+).

B.Eng. in Automation, Tianjin University 2021.08-2025.07

Main Courses: Introduction of Artificial Intelligence (95), Basis of Pattern Recognition (98), Image Processing Technologies (97), Basis of Optimization Technology (95), Autocontrol Theory (90).

Publication & Patent

1. **Rui Wang**, Jincheng Zhao, Yi Tang, Qingtao Zhang, Dayang Xie, Wei Dong*, Xiaoyan Li*. Association of small dense low-density lipoprotein cholesterol with longitudinal trajectories of cardio-vascular-kidney-metabolic comorbidity: A prospective cohort study based on multi-state Markov models. *Maturitas*, 2026. DOI: 10.1016/j.maturitas.2026.108950.
2. Jianing Li¹, **Rui Wang**¹, Zhou Yi, Jianhui Ni, Zhigang Zuo*, Yue Wang*. Comparing Deep Learning Models for Identifying Maxillary Transverse Deficiency from Intraoral Photographs. *International Dental Journal*. 2026;76(4):109689. DOI: 10.1016/j.identj.2026.109689.
3. **Rui Wang**, Xinyi Tang, Shuangming Yang*. Peak-driven Multi-scale Learning with a Mixing Mechanism with Spiky Dendrites. *The 5th China Symposium on Cognitive Computing and Hybrid intelligence*. Paper ID: CCHI2023-102443. 2023.09.30.
4. Donghui Li, **Rui Wang**. A Multi-Wheel Synchronous Control Method and System for Ship-Cleaning Machines Based on SMC-CCC-LESO. Chinese Patent.

Internships

Intern, Prof. Yinliang Yue's Group, Zhongguancun Laboratory 2026.01-05

- **Event-chain modeling and data construction:** built cross-platform brand-related event-chain datasets from Weibo, Bilibili, Douyin, and Xiaohongshu to characterize temporal evolution of public brand cognition under consecutive events.
- **Group cognition and polarization analysis:** analyzed cognitive divergence paths of different user groups by integrating user behavior features and interaction networks; quantified stance shifts, emotional accumulation, and polarization dynamics.
- **Cognitive security mechanism modeling:** investigated polarization mechanisms from event sequencing, group heterogeneity, and cognitive-object migration perspectives to support brand-related cognitive risk identification and early warning.

Research Assistant, Prof. Cheng Chen's Group, The University of Hong Kong 2025.12-2026.05

- **Multimodal modeling for survival prediction:** developed survival prediction models integrating radiology, pathology, genomic, and demographic data, targeting real-world clinical scenarios with incomplete modalities.
- **Robust learning under missing modalities:** designed a mean-vector-based multimodal fusion framework with modality dropout and feature reconstruction constraints to effectively leverage incomplete multimodal data.
- **Experimental validation and analysis:** conducted large-scale experiments and ablation studies on real medical datasets to evaluate model robustness and performance gains under missing-modality conditions.

Intern, Chinese People's Liberation Army General Hospital 2023.06-08, 2025.09-2026.01

- **Medical data analysis and visualization:** Calculated Annual Percentage Change (EAPC) and Age-Standardized Rate (ASR) using the Global Burden of Disease (GBD) database; conducted correlation analyses and generated visualizations including scatter plots, density plots, disease composition charts, and global trend maps.

- **ECG-based volume status assessment:** Performed feature engineering by integrating ECG signals, BNP/NT-proBNP biomarkers, and clinical variables; developed machine learning models (e.g., XGBoost) for predicting volume status and risk assessment in heart failure patients.
- **CKMM comorbidity evolution analysis:** Constructed comorbidity networks using multi-center clinical data to investigate the evolution patterns of Cardiovascular-Kidney-Metabolic (CKMM) diseases and developed predictive models for disease progression (first-author paper, JCR Q1).
- **Oral image-based intelligent diagnosis:** Built a dataset of frontal intraoral photographs paired with CBCT images and applied deep learning models (e.g., ResNet) to automatically identify Maxillary Transverse Deficiency (MTD) (co-first-author paper).

Research Assistant, Prof. Lingjie Kong's group, Tsinghua University

2024.01-03

- **Application of deep learning in liver cancer prognostic biomarker identification:** studied and successfully reproduced the PathFinder algorithm for biomarker recognition.
- **Visual cortex signal decoding:** reproduced the CEBRA algorithm to analyze decoding in mice during video viewing.
- **Brain data processing & prediction:** processed neural data from mice exposed to complex videos; used the DINO algorithm to compress videos into image representations and predict subsequent frames.

Research Projects

Longitudinal Trajectory Modeling of Cardiovascular-Kidney-Metabolic Multimorbidity Associated with sdLDL-C (First Author, JCR Q1)

2025.09-2026.04

- **Longitudinal Multi-state Modeling Design:** Constructed a multi-state Markov model based on the prospective CHARLS cohort to characterize disease progression trajectories from healthy state → single disease → cardiovascular-kidney-metabolic multimorbidity (CKMM).
- **Risk Association and Stage-specific Transition Analysis:** Integrated Cox proportional hazards models with transition intensity estimation to systematically evaluate the stage-dependent effects of sdLDL-C and LDL-C on disease progression probabilities.
- **Residual Risk Identification and Mechanism Assessment:** Conducted stratified and sensitivity analyses to validate the residual predictive value of sdLDL-C beyond LDL-C control, identifying its stage-specific role in early transition and multimorbidity progression.
- **Model Validation and Robustness Evaluation:** Performed multiple imputation, nonlinear relationship testing, and model robustness analyses to ensure statistical reliability and mechanistic interpretability of findings.

Deep Learning-Based Intelligent Diagnosis of Maxillary Transverse Deficiency (Co-First Author, JCR Q1)

2025.09-2026.05

- **Study Design and Dataset Construction:** Built an orthodontic imaging dataset consisting of 826 intraoral frontal photographs paired with CBCT scans. Used CBCT-derived UPA/YTA measurements as reference standards for sample labeling and established a standardized medical image preprocessing pipeline.
- **Model Development and Comparative Evaluation:** Implemented and systematically evaluated multiple deep learning architectures using PyTorch, including DenseNet, ResNet, EfficientNet, and MobileNet, to compare their performance in medical image classification tasks.
- **Statistical Validation and Generalization Assessment:** Employed 5-fold cross-validation to evaluate model robustness and calculated 95% confidence intervals using bootstrap resampling. Conducted DeLong tests to compare AUC differences between models, and performed independent validation on the HKU FD-Tooth external dataset ($n \approx 200$) to assess model generalizability.
- **Diagnostic Performance and Clinical Significance:** Achieved $>99\%$ classification accuracy under the YTA labeling scheme using only intraoral photographs without CBCT imaging, demonstrating strong potential for reducing radiation exposure, lowering diagnostic costs, and improving efficiency in orthodontic screening.

VideoTutor: An Adaptive Knowledge Bot for Enhanced Learning via RAG and LLM-Assisted Video Generation

2025.08-2026.05

- **System architecture design:** designed an intelligent teaching system centered on instructional PDFs, integrating document parsing, semantic retrieval, RAG-based question answering, and multimodal content generation for real-world educational scenarios.
- **Reliable QA with RAG:** implemented page-level semantic retrieval and source-grounded answer generation under the RAG framework to reduce LLM hallucinations and improve answer accuracy and interpretability.

- **End-to-end multimodal content generation:** built an automated pipeline from teaching PDFs to presentation scripts, TTS audio, instructional videos, and multilingual subtitles, significantly improving content production efficiency.
- **Human-in-the-loop system implementation:** introduced a teacher–student dual-role review mechanism for manual revision and quality control; implemented and validated the full system in Python.

Design of Speed Control System for Four-Wheel Drive Ship Washer - Excellent Graduation Project

2025.01-06

- **System Architecture:** Designed overall scheme and key hardware based on eCore640S-I industrial controller.
- **Control Strategy:** Developed a composite synchronous control method integrating sliding mode control, cross-coupling, and LESO to address hydraulic nonlinearity and multi-wheel coordination, improving speed accuracy and robustness.
- **Modeling & Simulation:** Built system models in MATLAB/Simulink to verify steady-state and synchronization performance under disturbances.
- **Control Implementation:** Programmed in CoDeSys using state machines for logic control, enabling start/stop, forward/reverse, and speed scheduling in a closed-loop system.
- **Intellectual Property:** Co-inventor (2nd inventor) of a Chinese invention patent related to the proposed multi-wheel synchronous control method (first inventor: supervisor).

Brain-Inspired Vision-Based Autonomous Driving System - Core Member

2023.03-2024.04

- **Hardware & Layout:** Designed system block diagram and PCB in Altium Designer; completed drilling, encoder installation, circuit board layout, and camera assembly.
- **Software Development:** Programmed hardware drivers, speed control, and camera-based image processing for trajectory recognition and autonomous control.
- **Conference Paper:** First author of extended abstract “Multi-Scale Learning Driven by Spiking-Dendritic Hybrid Mechanism”, Paper ID: CCHI2023-102443.

Competitions

the 2nd prize in the North China Area, the 2nd University Electrical and Electronic Engineering Innovation Competition

2023.03-09

Built integrated energy optimal scheduling model for cold-heat-electric-gas systems for typical seasonal variability and source-load uncertainty.

Activities & Volunteering Experiences

Deputy Leader & Section Leader of Peiyang Wind Orchestra

2021.08-2025.07

Performance and Event Coordination: Participated in multiple on- and off-campus performances, including at Tianjin Grand Theatre and Tianjin Ren'ai College; awarded “Model Orchestra” at the Beijing International Music Festival and First Prize at the Tianjin Art Exhibition; organized and coordinated over ten performances.

Other Information

Skills: R Studio, Python, and MatLab

Hobbies: percussion music (the Chinese Conservatory of Music 10th level in drum set), and debate.